Cyclic Incremental Latin $n \times n$ Squares encoding Generalized (n, c)-Coins Squares for $0 \leq c \leq n$

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Abstract

For every natural number $n \notin \{0, 2, 3, 4, 6\}$, there exists a permutation $q : \operatorname{Fin} n \to \operatorname{Fin} n$ such that $L_{i,j} = (i + q(j)) \bmod n$ is an *Cyclic Incremental Latin Square*, encoding (n, c)-Coins Squares for $0 \leq c \leq n$.

An (n, c)-Coins Square for $0 \leq 1 \leq n$ corresponds to a solution of n-Queens Problem.

We have an existence proof in `Lean` and construction code in `Haskell`, `Julia` and `Lean`.

See e.g. MO#510135, PSE#138939, MO#516225 and GitHub.

The proof uses the *Injective-Majorization Principle* that, in combination with the *Pigeon Hole Principle*, replaces intensive checking inequalities on each diagonal to the existence of a single function $f(i, j) \leq L_{i,j}$ that is injective on every diagonal.

The proof uses four families covering almost all natural number modulo 6 classes

- for $n \equiv 1, 5 \pmod 6$ (with $n \geq 1$)
- for $n \equiv 2, 4 \pmod 6$ (with $n \geq 10$)
- for $n \equiv 0 \pmod 6$ (with $n \geq 12$)
- for $n \equiv 3 \pmod 6$ (with $n \geq 27$)

For all four families a Cyclic Incremental Latin Square can be constructed. Constructing a Cyclic Incremental Latin Square for the four uncovered natural numbers, $n \in \{8, 9, 15, 21\}$, is handled separately. Likewise, constructing a (non-Cyclic) Incremental Latin Square for the two remaining natural numbers, $n \in \{4, 6\}$, is also handled separately.

1 Cyclic Incremental Latin Squares

1.1 From (n, c)-Coins Squares to Cyclic Incremental Latin Squares

(n, c)-Coins Squares are subsets $S_c \subseteq \{0, \ldots, n-1\}^2$ of nc cells of an $n \times n$ square such that every row, column, sum-diagonal ($i + j = s$), and difference-diagonal ($i - j = d$) contains at most c cells of S_c.

Clearly, for $c = 1$, they correspond to solutions of n-Queens Problems.

The construction of Le, Li and Wang [1] constructs S_c's separately for each c.

1.2 The Conjecture

Every permutation $q : \operatorname{Fin} n \to \operatorname{Fin} n$ defines a *Cyclic Latin Square*:

$$L_{i,j} = (i + q(j)) \bmod n \qquad (0 \leq i, j < n).$$

The conjecture is that for all $n > 0$ except $n \in \{2, 3, 4, 6\}$, there exists a column permutation q such that $L_{i,j} = (i + q(j)) \bmod n$ is a Cyclic Incremental Latin Square. For $n \in \{2, 3\}$, the n-queens problem has no solution, and, for $n \in \{4, 6\}$, exhaustive finite search confirms that no Cyclic Incremental Latin Square satisfies the diagonal bounds.

Theorem 1.1 (Cyclic Incremental Latin Square Conjecture)**.** *For every integer $n > 0$ and $n \notin \{2, 3, 4, 6\}$, there exists a bijection $q : \operatorname{Fin} n \to \operatorname{Fin} n$ such that the Cyclic Incremental Latin Square*

$$L_{i,j} = (i + q(j)) \bmod n$$

satisfies:

(i) For every $i \in \operatorname{Fin} n$, the row map $j \mapsto L_{i,j}$ is bijective.
(ii) For every $j \in \operatorname{Fin} n$, the column map $i \mapsto L_{i,j}$ is bijective.
(iii) For every $0 \le k < n$ and every $s \in \mathbb{N}$,

$$|\{(i, j) \in \operatorname{Fin} n \times \operatorname{Fin} n \mid i + j = s \ \wedge \ L_{i,j} \le k\}| \le k + 1.$$

(iv) For every $0 \le k < n$ and every $d \in \mathbb{Z}$,

$$|\{(i, j) \in \operatorname{Fin} n \times \operatorname{Fin} n \mid i - j = d \ \wedge \ L_{i,j} \le k\}| \le k + 1.$$

2 Injective Majorization

Instead of direct cardinality counting of

$$|\{(i, j) \in \operatorname{Fin} n \times \operatorname{Fin} n \mid i + j = s \ \wedge \ L_{i,j} \le k\}| \le k + 1.$$

and

$$|\{(i, j) \in \operatorname{Fin} n \times \operatorname{Fin} n \mid i - j = d \ \wedge \ L_{i,j} \le k\}| \le k + 1.$$

which would computationally explode for large n, we use a simple combinatorial principle that completely eliminates cardinality counting and reduces the problem for *all* thresholds $0 \le k < n$ simultaneously to constructing a single dominated injection.

2.1 The Injective-Majorization Principle

Let $D \subseteq \operatorname{Fin} n \times \operatorname{Fin} n$ be any diagonal (either a sum-diagonal $i + j = s$ or a difference-diagonal $i - j = d$), and let $v(i, j) = (i + q(j)) \bmod n \in \{0, \ldots, n - 1\}$ be the square value at cell (i, j).

Lemma 2.1 (Injective-Majorization Principle)**.** *If there exists a function $f : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ such that:*

(1) Majorization: $f(i, j) \le v(i, j)$ for all $0 \le i, j < n$,
(2) Diagonal Injectivity: f is injective along D.

Then for every threshold $k \in \mathbb{N}$, the set $D_{\le k} = \{(i, j) \in D \mid v(i, j) \le k\}$ has cardinality at most $k + 1$.

Proof. For every cell $(i, j) \in D_{\le k}$, condition (1) gives

$$0 \le f(i, j) \le v(i, j) \le k,$$

so $f(i, j) \in \{0, \ldots, k\}$. By condition (2), the restriction of f to $D_{\le k}$ is an injective map into $\{0, \ldots, k\}$. By the Pigeonhole Principle,

$$|D_{\le k}| \le |\{0, \ldots, k\}| = k + 1. \qquad \square$$

Remark 2.2 (Geometric Meaning of f). Along a diagonal D, the square values $v(i, j)$ are *almost* distinct, except for a small number of pairs $(i_1, j_1), (i_2, j_2) \in D$ caused by jumps between piecewise-linear blocks of $q(j)$ or modular wrap-around at n. Whenever two cells on D share the same value $v(i_1, j_1) = v(i_2, j_2) = v$, one of the two cells is allowed to keep its value, say $f(i_1, j_1) = v$ while the other is changed to an unused value $f(i_2, j_2) < v$ that does not appear anywhere else along D. Because $f(i_2, j_2) < v(i_2, j_2)$ on the changed cell, the changed cell steals a slot in $\{0, \ldots, f(i_2, j_2)\} \subseteq \{0, \ldots, v(i_2, j_2)\}$ that was left vacant on D!

References

[1] Mao-Hua Le, Wen Li, and Edward T. Wang, *A generalization of the n-queen problem*, Journal of Systems Science and Mathematical Sciences **3** (1990), no. 2, 183–192, https://sysmath.cjoe.ac.cn/jssc/CN/Y1990/V3/I2/183.